

Wentworth Liu

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PROFESSIONAL SUMMARY

Data Scientist focused on operational analytics for physical EV assets. I use Python, SQL, statistical modelling and applied ML to turn high-volume battery data into monitored risk, prioritised investigations and targeted interventions-from scalable data pipelines and model evaluation through alerts, daily work orders and feedback-driven improvement. Relocating to London; authorised to work in the UK without sponsorship.

RELEVANT IMPACT

Million+ connected vehicles | TB-scale battery data | 15k+ features | 93% Recall@2% | 30+ fault categories | F1 > 0.90 | ~85% faster iteration | MLflow | Spark | APIs

PROFESSIONAL EXPERIENCE

Li Auto July 2024-Present

Data Scientist, Battery Intelligence Department

Beijing, China

01 / Fleet Risk Monitoring & Prioritisation

Across million+ connected vehicles and multiple battery platforms, developed an explainable thermal-runaway risk-ranking system to prioritise the highest-risk cases. Used fleet battery data and risk-driver analysis to build and retrospectively validate the model; achieved 93% Recall@2%. Deployed cloud infrastructure, MLflow, API services and alerting to automate vehicle-level scoring and continuous monitoring.

02 / Battery Degradation Decision Support

For SOH-degradation risk across vehicle models, identified high-risk cohorts and actionable drivers through quasi-causal analysis. Productised the workflow to detect abnormal-lifetime batteries, classify degradation patterns and generate tailored intervention strategies. The resulting strategies were projected to reduce warranty exposure by tens of millions of RMB for one model.

03 / TB-Scale Feature & Data Platform

To make battery data reusable and reliable for developers, designed and rolled out an internal feature platform spanning TB-scale data and 15k+ features. Built Spark-based pipelines with custom event definitions, multi-operator aggregations and cross-platform normalisation; added feature querying, data-quality monitoring, definition configuration, asset management and evaluation. Improved feature-iteration efficiency by ~85%.

04 / Closed-Loop Fault Classification

Built a hybrid rule-based/statistical classifier for 30+ battery-quality fault categories and automated training, evaluation and daily work-order reporting. Added data and model quality checks, then fed after-sales human corrections back into the workflow for continuous improvement. Reached Weighted F1 > 0.90 while creating a reliable operating loop for fault investigation.

ByteDance May 2023-September 2023

Data Analytics Intern, Douyin Market Strategy

Beijing, China

- Integrated consumption, dissemination and search signals into an IP-popularity measurement system; added automated pipeline monitoring to address upstream data instability and support recurring content-investment and promotional decisions.
- Redesigned performance metrics and a multi-dimensional content taxonomy for Qishui Music; used funnel segmentation and KOL audience-saturation modelling to diagnose new-user conversion decline and inform content selection and marketing-resource allocation.

Dongwu Securities December 2021-March 2022

Equity Research Intern, Mechanical Sector

China

- Combined company fundamentals, competitive analysis and market dynamics with stock-price modelling to produce structured short-, medium- and long-term investment recommendations.

EDUCATION

MSc, Computational Materials Science Nanjing University 2021-2024

Developed Skeaf, a 3D Fermi-surface visualisation tool using HDBSCAN clustering and quantum-oscillation data; first author, Physical Review B (2024); co-author, Physical Review B (2024) and Physical Review Materials (2023); patent CN115795266A.

BSc, Materials Science & Engineering Nanjing University 2017-2021

Outstanding Graduate | GPA 4.6/5.0 | Top 1% | National Scholarship (top 0.15%) | MCM/ICM 2020 Honorable Mention (Team Captain).

TECHNICAL SKILLS

Programming & Workflow: Python, SQL/HiveSQL, C++, MATLAB, Git, GitHub

Operational Analytics & Data: Spark, Power BI, Multi-source Data Integration, Data Pipelines, Risk Analytics, Monitoring & Decision Support, Data Productisation, Visualisation, Linux, Hadoop

ML & Statistics: MLflow, PyTorch, LightGBM, scikit-learn, Machine Learning, Statistical Modelling, Causal Inference, Predictive Modelling, Feature Engineering, Time Series, NLP

Cloud & Delivery: AWS, GCP, Docker, CI/CD, APIs, Automation

Domain & Languages: Battery Intelligence, EV Analytics, Scientific Computing; English (TOEFL 104), Mandarin/Cantonese (native), Japanese (JLPT N1)